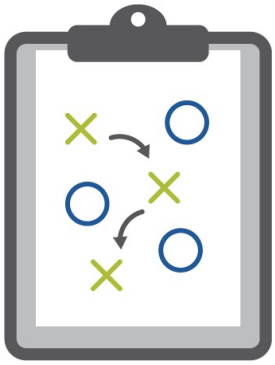
Activity: Experience teams – Purpose

Duration 5 min

Purpose

- Experience how an Agile Team functions
- Get as many balls through the group as possible within two minutes
- After two minutes, the group is allowed an additional minute (one-minute Retrospective) to discuss the process and how it could be improved. The game is played a total of three times. (three two-minute Iterations).



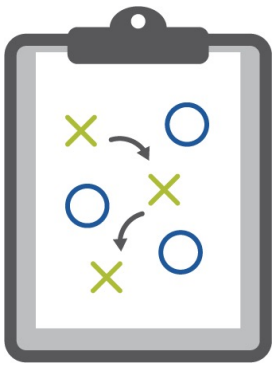
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Activity: Experience teams – Rules

Duration 5 min

Rules

- Everyone is part of one big group
- Each ball must have air-time
- Each ball must be touched at least once by everyone
- Balls cannot be passed to your direct neighbor (to your immediate left or right)
- Each ball must return to the same person who introduced it into the system
- There are a total of three Iterations



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Activity: Experience teams

Prepare 15 min | Share 5 min

- **Step 1:** Organize into small groups.
- **Step 2:** As a group, estimate how many balls you think you can process in two minutes.
- **Step 3:** Run the two-minute Iteration following the rules (get as many balls through the group as possible)
- **Step 4:** Take one minute (run the one-minute Retrospective) to discuss how you can improve the process

Note: To get credit, you must provide an estimate for the number of balls you think you can process before each Iteration.

- **Step 5:** Summarize your experience as a group:
  - What were some of the challenges?
  - How were you able to improve the process with each Iteration?

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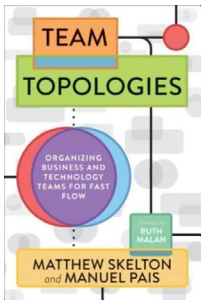
Teams on the ART are organized for flow

Stream-aligned team – organized around the flow of work and has the ability to deliver value directly to the Customer or end user.

Complicated subsystem team – organized around specific subsystems that require deep specialty skills and expertise.

Platform team – organized around the development and support of platforms that provide services to other teams.

Enabling team – organized to assist other teams with specialized capabilities and help them become proficient in new technologies.



More information in the Advanced Topic Article:
https://www.scaledagileframework.com/organizing-agile-teams-and-arts-team-topologies-at-scale/

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Applying the four topologies

- **Stream-aligned Teams (best)**
  - By product, Solution, or service
  - By Customer or market segment
  - By Solution feature areas
  - By steps in the Customer journey
  - By value streamlets
  - New product innovation
- **Complicated Subsystem Teams**
  - Highly specialised system components
  - Safety critical systems elements
  - Specialty algorithm or business rules
  - Part of a cyber-physical system

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Applying the four topologies - continued

- **Platform Teams**
  - Sets of services consumed by other Teams
- **Enabling Teams**
  - DevOps implementation
  - Automated testing
  - Continuous integration and build tooling
  - Engineering quality practices
  - Security Environments and configuration

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Activity: Building your team

Prepare

10 min

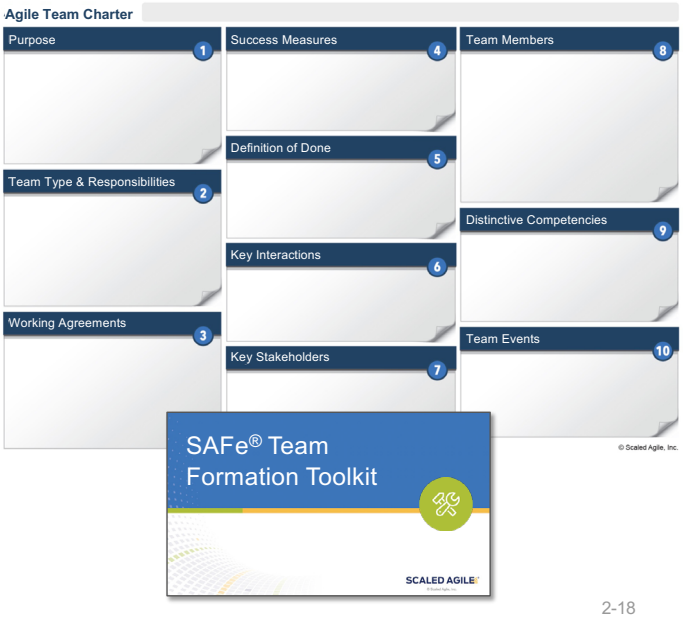
- **Step 1:** As a group, discuss each person's responsibilities and skill sets.
- **Step 2:** Create a group name. Note: Names should not be the names of components, subsystems, or Feature areas. Instead, create a fun name, a mascot, and a cheer.
- **Step 3:** Discuss your role as one of the four team topologies.
- **Step 4:** Discuss what your group is responsible for and what other things you can do.
- **Step 5:** Prepare a short presentation about your group (name, role on the train, and special skills that other groups should know about).

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Facilitate an Agile Team Charter workshop

Part of the SAFe Team Formation Toolkit.

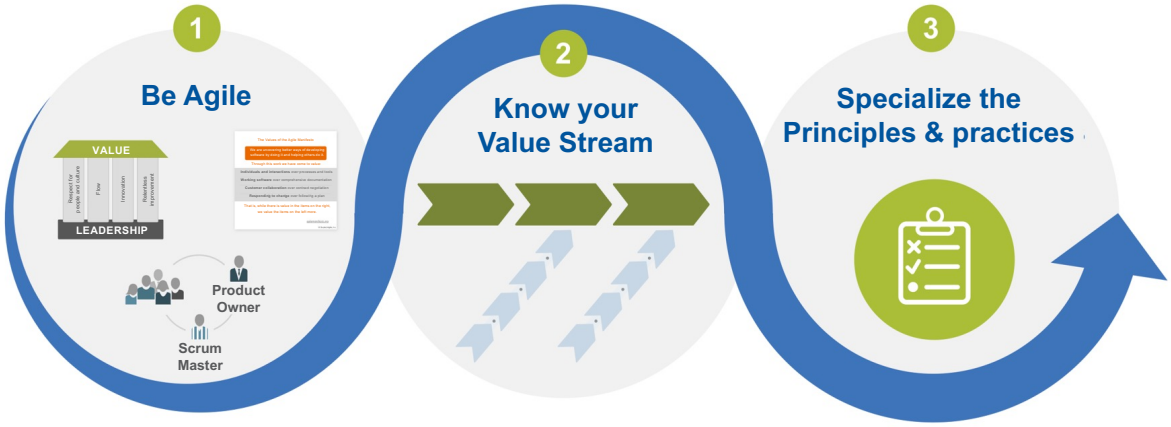
- The Agile Team Charter helps teams to clearly define their purpose, responsibilities and success criteria amongst other critical elements.
- The process of completing the Agile Team Charter provides the opportunity for teams to discuss and reflect both on how they want to work together and with other teams on the ART.
- For each of the 10 boxes on the charter an interactive exercise is included to generate discussion and the required output.



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Extend into the business with Agile business teams



Agile Team maturity cycle

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2.2 The Scrum Master and Product Owner roles

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Agile Teams have two speciality roles

Scrum Master

- Coaches the Agile Team in self-management
- Helps the team focus on creating increments of value each iteration
- Facilitates the removal of impediments to the team's progress
- Ensures that all team events take place, are productive and kept within the timebox

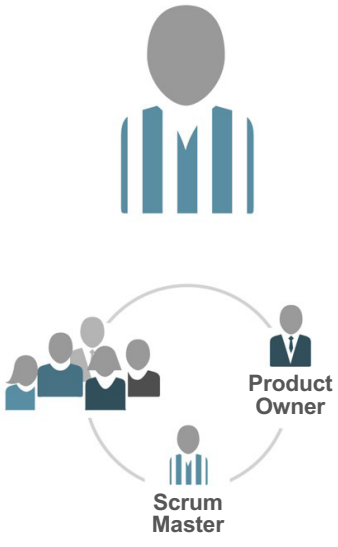
Product Owner

- Contributes to the Vision and Roadmap
- Acts as the Customer for team questions
- Creates, clearly communicates and accepts Stories
- Prioritizes the Team Backlog

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The Scrum Master in the Enterprise

- Coordinates with other Product Owners, the System Team, and Shared Services in the PI Planning meetings
- Works with the above teams throughout each Iteration and PI
- Coordinates with other Scrum Masters and the Release Train Engineer in the Scrum of Scrums
- Helps team understand and operate within its capacity
- Helps team operate under architectural and portfolio governance, system-level integration, and System Demos
- Fosters team adoption of Agile technical practices

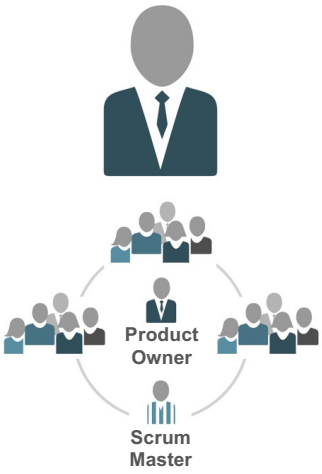


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The Product Owner in the Enterprise

- Establishes the sequence of backlog items based on program priorities, events, and dependencies with other teams
- Operates as part of an extended Product Management Team
- Understands how the Enterprise backlog structure operates with Epics, Capabilities, Features, and Stories
- Work with Product Management to plan Program Increments (PI)
- Uses PI Objectives and Iteration Goals to communicate with management
- Coordinates with other Product Owners, the System Team, and Shared Services in the PI Planning meetings
- Works with other Product Owners and the Product Management team throughout each Iteration and PI



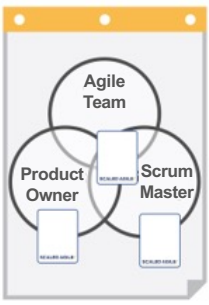
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Activity: Scrum roles and responsibilities

Prepare 8 min

Share 2 min

- **Step 1:** With your group, draw the following Venn diagram on a flipchart sheet
- **Step 2:** Review the responsibility cards
- **Step 3:** Place them either in the role or at an intersection of the Venn diagram
- **Step 4:** Present your Venn diagram to the class



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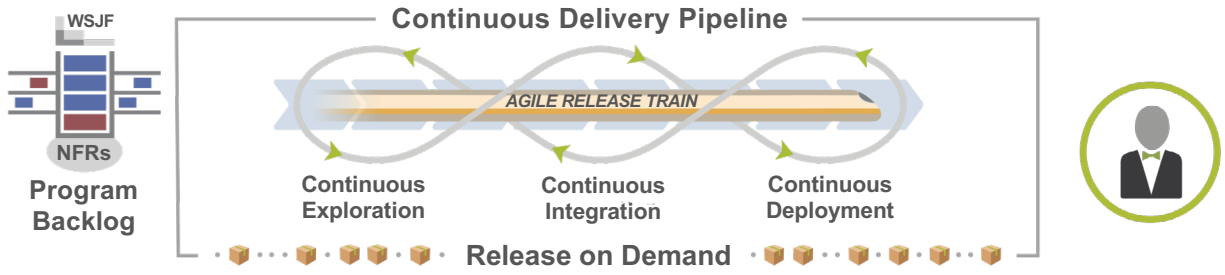
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2.3 Organizing Agile Release Trains

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Agile Release Trains (ARTs)

- A virtual organization of 5 – 12 teams (50 – 125+ individuals)
- Synchronized on a common cadence, a Program Increment (PI)
- Aligned to a common mission via a single Program Backlog



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Roles on the Agile Release Train

Release Train Engineer acts as the chief Scrum Master for the train.

Product Management owns, defines, and prioritizes the Program Backlog.

System Architect/Engineering provides architectural guidance and technical enablement to the teams on the train.

System team provides processes and tools to integrate and evaluate assets early and often.

Business Owners are key stakeholders on the Agile Release Train.



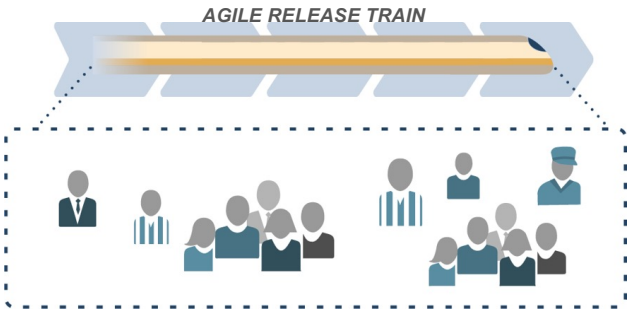
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2-27

Activity: Know the people on the train

Share 20 min

- **Step 1:** The RTE introduces themself
- **Step 2:** The RTE presents the main players on the train:
  - Product Management
  - System Architect/Engineering
  - Lean UX
  - Shared Services
- **Step 3:** Each group presents itself (name, area of responsibility, special skills)



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Action Plan: Building an Agile Team

Prepare 5 min

Share 3 min

- **Step 1:** As a group, brainstorm one to three actions you could take to improve in any areas related to this lesson
- **Step 2:** Individually write down at least one improvement item
- **Step 3**: Share one item you discussed as a group and one item you individually wrote in your Action Plan



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Building an Agile Team

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Lesson review

In this lesson you:

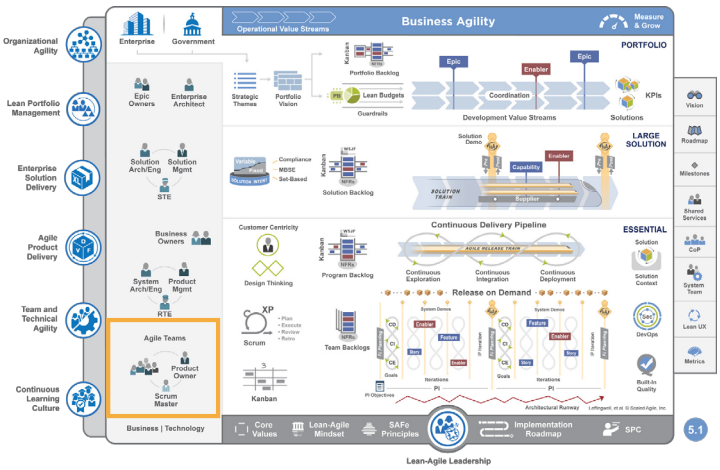
- Built your team and learned about their roles
- Explored the roles of the Scrum Master and the Product Owner
- Identified the people and teams on the train, and learned about their roles

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Agile Teams" https://www.scaledagileframework.com/agile-teams/
- "Product Owner" https://www.scaledagileframework.com/product-owner/
- "Scrum Master" https://www.scaledagileframework.com/scrum-master/
- "Organizing Agile Teams and ARTs: Team Topologies at Scale" https://www.scaledagileframework.com/organizing-agile-teams-and-arts-team-topologies-at-scale/



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Continue your SAFe journey with the following resources:

Watch this three-minute video, *Welcome to Your Scrum Team,* for an overview of what to expect in your first days on a Scrum team, and the role each team member plays in executing as a team. **https://bit.ly/Video-WelcometoYourScrumTeam**	Download and use the "Team Formation Toolkit (5.1)" to build your Agile Team charter and clearly define your purpose, responsibilities and success criteria, and other critical elements necessary for your new team to flourish. **https://bit.ly/Community-ToolkitsandTemplates**
Use the "Measure and Grow Workshop Toolkit (5.1)" to run a workshop to identify growth opportunities for your team as you work toward mastering SAFe and Business Agility. **https://bit.ly/Community-ToolkitsandTemplates**	Share the "Essential SAFe Toolkit (5.1)" with your team to help ensure a shared understanding of the basic building blocks needed for a successful SAFe implementation. **https://bit.ly/Community-ToolkitsandTemplates**
Facilitate an exercise using the "Experience Teams!" SAFe Collaborate template to practice working as a team and using retrospectives to improve team performance. **https://bit.ly/Template-ExperienceTeams**	Use the "Agile Team Charter Virtual Workshop" in SAFe Collaborate to help a remote or distributed team capture the ideas and decisions from the "Team Formation Toolkit (5.1)." **https://bit.ly/Template-SAFeAgileTeamCharter**

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 3

Planning the Iteration

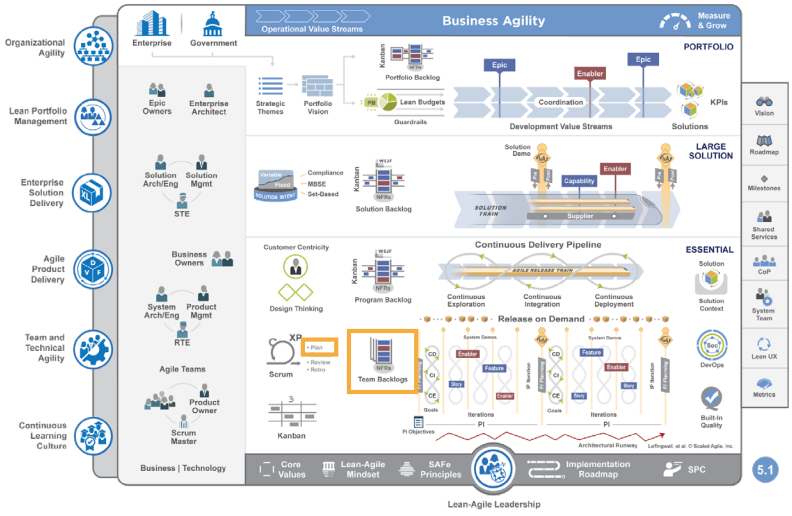
SAFe® Course - Attending this course gives learners access to the SAFe® Practitioner exam and related preparation materials.

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Lesson Topics

3.1 Preparing the Team Backlog

3.2 Planning the Iteration



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Learning objectives

At the end of this lesson, you should be able to:

- Identify the qualities of well-written Features and Stories
- Prepare your backlog by breaking down Features into Stories
- Discuss the key considerations for sequencing work
- Apply estimation poker for fast, relative estimation
- Explain how to conduct an Iteration Planning event
- Explore Kanban as a team method for visualizing workflow

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3.1 Preparing the Team Backlog

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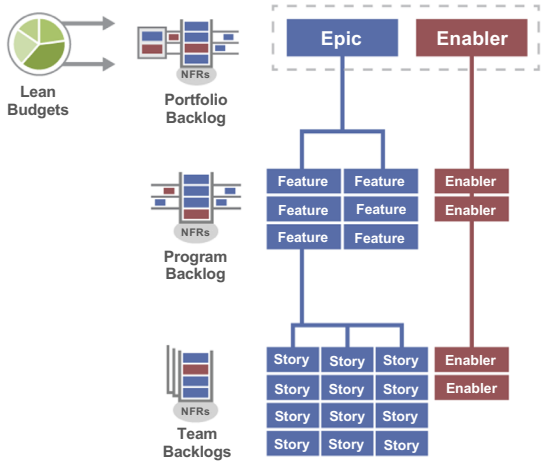
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Define Features for the Program Backlog

Features are services that fulfill stakeholder needs.

- Feature is an industry-standard term familiar to marketing and Product Management
- Describe Features with a short phrase and a benefit hypothesis that clearly expresses their value
- Identify, prioritize, estimate, and maintain Features in the Program Backlog



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Features represent the work for the Agile Release Train

- The Feature benefit hypothesis justifies development cost and provides business perspective for decision-making
- Features contain acceptance criteria typically defined during Program Backlog refinement
- They reflect functional and non-functional requirements
- Features fit in one PI

In-service software update

Benefit hypothesis
Significantly reduced planned downtime

Acceptance criteria
1. Nonstop routing availability
2. Automatic and manual update support
3. Rollback capability
4. Support through existing admin tools
5. All enabled services are running after the update

Example Feature

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Example Features

Software Example

Multi-factor authentication

Benefit hypothesis

Enhanced user security will reduce risk of a system data breach

Acceptance criteria

1. USB tokens as a first layer
2. Password authentication second layer
3. Multiple tokens on a single device
4. User activity log reflecting both authentication factors
5. Data breach tests pass

Business Example

Create GDPR Incident Response Plan

Benefit hypothesis

Organizational readiness to quickly respond to incidents

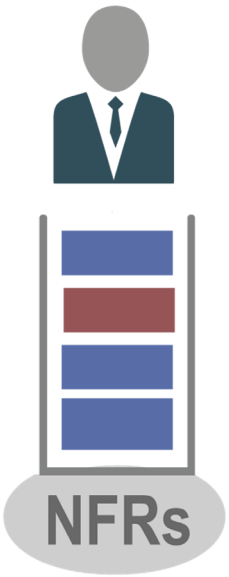
Acceptance criteria

1. Incident response plan is fully documented
2. Incident response plan is reviewed and approved by PO
3. Incident response is compliant with legal requirements

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The Team Backlog

- Contains all the work for the team
- Created by the Product Owner and the team
- Prioritized by the Product Owner
- Contains User and Enabler Stories
  - User Stories provide Customers with value
  - Enabler Stories build the infrastructure and architecture that makes User Stories possible
- Stories for the next Iteration are more detailed than Stories for later Iterations
- Nonfunctional requirements (NFRs) are a constraint on the backlog



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User Stories

- User Stories are short descriptions of a small piece of desired functionality, written in the user's language
- The recommended form of expression is the user-voice form, as follows:
  As a (user role), I want to (activity), so that (business value)

As a driver, I want to limit the amount of money before I fuel so that I can control my expenditure

As a driver, I want to get a receipt after fueling so that I can expense the purchase.

As the Finance Department, we want to print receipts only for drivers who request them so that we save on paper

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3-9

Using personas to better understand users

Personas are detailed fictional characters acting as a representative user.



Jane - Mileage sensitive
- Law-abiding driver
- Obeys all traffic signs
- Wants to save on gas



Bob - Time sensitive
- Impatient driver
- Ignores traffic signs if they slow him down

As Jane, I want to travel at the legal limit and operate in an energy saving manner so that I do not get a ticket and I save money

As Bob, I want to travel at the maximum speed the roadway and my vehicle safely allows so that I arrive quickly

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3-10

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INVEST in a good Story

I	**Independent**	Write Stories that can be developed separately
N	**Negotiable**	Write Stories in which scope can be negotiated
V	**Valuable**	Write Stories that are valuable to the Customer
E	**Estimable**	Write Stories that can be estimated
S	**Small**	Write Stories that can fit in an Iteration
T	**Testable**	Write Stories that are testable

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Writing good Stories: The 3Cs

Card

Written on a card or in the tool and may annotate with notes

As a spouse, I want a clean garage so that I can park my car and not trip on my way to the door.

Conversation

The details are in a conversation with the Product Owner

What about the bikes?

Oh yeah, hang the bikes

Confirmation

Acceptance criteria confirm the Story correctness

- Tools have been put away
- Items on the floor have been returned to the proper shelf
- Bikes have been hung

Source: 3Cs coined by Ron Jeffries

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Enabler Stories

Enabler Stories build the groundwork for future User Stories. There are four types of Enabler Stories:

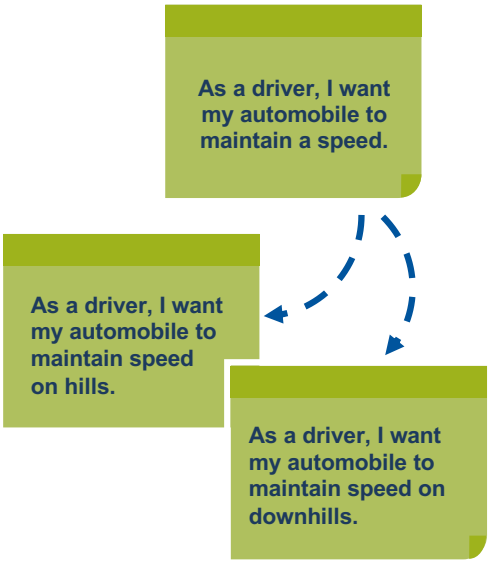
- **Infrastructure:** Build development and testing frameworks that enable a faster and more efficient development process
- **Architecture:** Build the Architectural Runway, which enables smoother and faster development
- **Exploration:** Build understanding of what is needed by the Customer to understand prospective Solutions and evaluate alternatives
- **Compliance:** Facilitate specific activities such as verification and validation, documentation, signoffs, regulatory submissions, and approvals

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Splitting Stories

- **In support of small batches for flow, decrease size to minimum:**
  - Split Stories into essential and non-essential parts and eliminate the non-essential
  - Ensure you have something releasable
- **In support of feedback:**
  - Deploy small Stories to get technical/user feedback quickly (maximize feedback)
- **In support of Iteration Planning:**

  Split Stories so they fit into an Iteration



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Apply some common splitting techniques

Splitting techniques:

- Business rule variations (e.g. single variation, then remainder)
- Workflow steps (for multi-step stories)
- Simple/complex (e.g. search for single word, then for phrases)
- Scenarios (e.g. use case exceptions)



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Activity: Break Features into Stories

Duration 15 min

Work with your group to break Features from the Program Backlog into Stories.

- **Step 1:** Select one Feature from the Program Backlog (your own or use the example provided in your workbook)
- **Step 2:** As a group, break the Feature into Stories in a way that they still retain business value
- **Step 3:** Write these User Stories in the User Story format:
  - **As a (user role), I want (activity) so that (business value).**



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Program Backlog Example Features

Feature: Flexible search

Benefit: Users will have a flexible, easy-to-use search capability to locate books.

Description: Search by author, title, or genre from a single search field. Mispelling subsitutions (i.e., "Did you mean ...?"). Present results as per-match algorithm.

Feature: Shopping Cart

Benefit: Users can manage items in a shopping cart for immediate or future purchase.

Description: Users can easily access their cart from any page, view the same infromation displayed in the book list, change the quantity, remove it from their cart, or save it for later. A subtotal for all items in their shopping cart should be displayed at the bottom. Items saved for later should appear below that.

Feature: Purchase by credit card

Benefit: Users can purchase products from us (as soon as implemented–only beta up until then)

Description: Users can select from their preferred credit card and shipping address as defined in their profile or add new ones. Visa, mastercard, Discover, and Diners Club ar required. American Express is optional. Must be PCI compliant.

Feature: Shipping method selection

Benefit: Users can select a shipping method based on cost, delivery speed, and carrier.

Description: Users can select a shipping method based on the price, delivery speed, and estimated delivery date for all major carriers (USPS, UPS, and FedEx).

Feature: Profile management

Benefit: Users can create and maintain their profiles rather than enter in their information each time they order.

Description: Users can manage their login credentials (ID, password), personal information (name, email address, home address), nickname for book rating and commenting, credit card information (multiple), and shipping address (multiple). Physical addresses, email addresses, and credit card info should be verified as valid. Passwords must meet current security standards.

Feature: Book detail

Benefit: Users can see informative and enticing details about a book.

Description: Display book name, book cover (which can be enlarged when clicked), author and bio, book description, genre, publishing info (publisher, release date, etc.), bok rating, and comments. Hyperlink author's name to a list of other books by the same author.

Feature: Book list sorting

Benefit: Users can sort a list of books in a number of ways to more easily find what they are looking for.

Description: Sort by book title, author, price, book rating, and release date. Allow for users to select the number of search results to appear on each page.

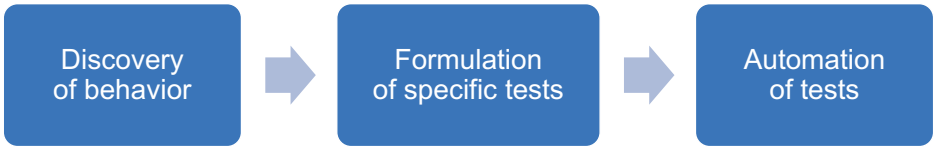
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Behavior-driven development: From ambiguity to precision

- Behavior is often first described in general terms, which can be ambiguous
- Specific examples of behavior provide better understanding
- The examples can directly become tests, or they can lead to specific behaviors which then are transformed into tests

Example



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Acceptance criteria

- Acceptance criteria provide the details of the Story from a testing point of view
- They are created by the Agile Team
- Can be written in the 'Given-When-Then' format

As a driver, **I want** to limit the amount of money before I fuel **so that** I can control my expenditure.

Acceptance Criteria:

1. **Given** that the driver indicated a maximum amount of money
   When the fuel cost reaches the amount
   Then the fueling process stops automatically
2. …

As a driver, **I want** to get a receipt after fueling so that I can expense the purchase.

Acceptance Criteria:

1. **Given** that the fueling is over
   When the driver asked for the receipt
   Then it is printed and includes:
   amount fueled, amount paid, tax, vehicle number, date, time

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Activity: Write acceptance criteria

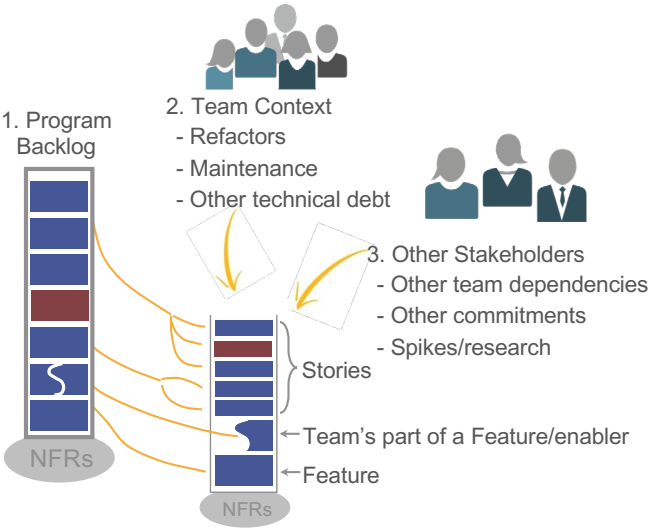
Prepare 10 min | Share 10 min

- **Step 1:** With your group, write acceptance criteria in the Given-When-Then format for three User Stories you have created.
- **Step 2:** Make sure the acceptance criteria are testable.
- **Step 3:** Discuss with your group:
  - Did writing acceptance criteria in the Given-When-Then format identify the need for any additional details?
- **Step 4:** Be prepared to share with the class.

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Sequencing Stories

- The Product Owner and the team sequence stories based on:
  - Priorities inherited from the Program Backlog
  - Capacity allocations for defects, maintenance, and refactors
  - Dependencies with other stories, teams, events, Milestones and releases.
- Initial sequencing happens during PI Planning
- Adjustments happen at Iteration boundaries



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3.2 Planning the Iteration

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Plan and commit with Iteration Planning

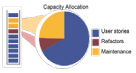
Purpose	Define and commit to what will be build in the Iteration
Process	• The Product Owner defines what • The team defines how and how much • Four hours max
Result	Iteration Goals and Backlog of the team's commitment
Reciprocal commitment	• Team commits to delivering specific value • Business stakeholder commit to leaving priorities unchanged during the Iteration



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Iteration Planning flow

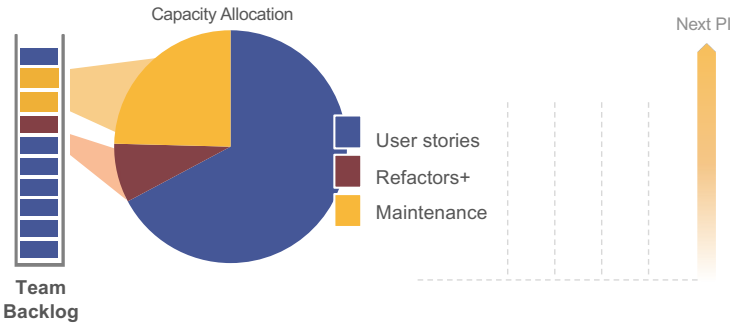
1. Establishing capacity 
2. Story analysis and estimating 
3. Detailing Stories 
4. Developing Iteration goals 
5. Committing to Iteration goals 

Iteration Planning

- **Timebox:** Four hours or less
- This event is **by** and **for** the team
- SMEs may attend as required

Capacity allocation for a healthy balance

- By having capacity allocation defined, the Product Owner doesn't need to prioritize unlike things against each other
- Once the capacity allocation is set, the PO and team can prioritize like things against each other



Capacity allocation

- Helps alleviate velocity degradation due to technical debt
- Keeps existing customers happy with bug fixes and enhancements
- Can change at Iteration or PI boundaries

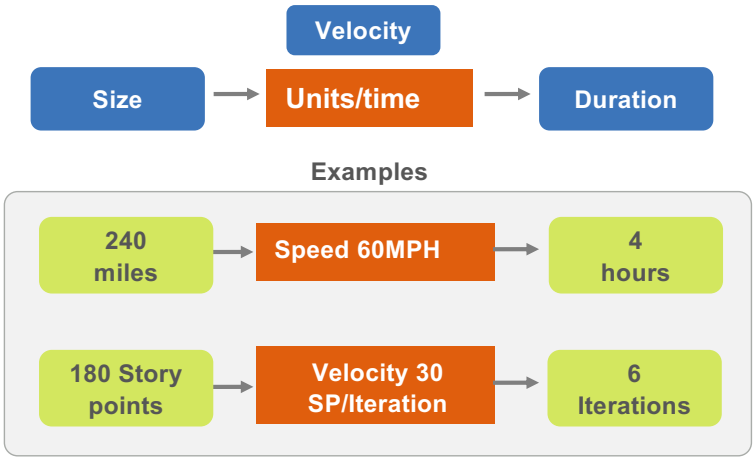
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Using size to estimate duration

Establish velocity by looking at the average output of the last Iterations.

Definition of Velocity

Velocity is the number of points of stories accepted in the Iteration. Make sure to always use the average velocity for the most recent Iterations.



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Story analysis and estimation

- The Product Owner presents Stories in order of priority
- Each Story
  - Is discussed and analyzed by the team
  - Has its acceptance criteria refined
  - Is estimated
- The process continues until the total Story points of the estimated Stories has reached the capacity of the team

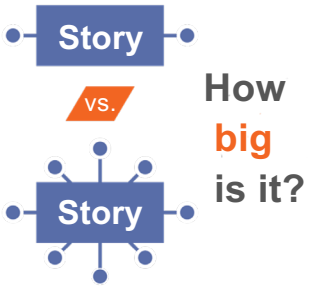
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Estimate Stories with relative Story points

- A Story point is a singular number that represents:
  - Volume: How much is there?
  - Complexity: How hard is it?
  - Knowledge: What do we know?
  - Uncertainty: What's not known?
- Story points are relative. They are not connected to any specific unit of measure.
  - An 8-point Story should take four times longer than a 2-point story to complete
  - Typically, a 1-point story would take one day to develop and test



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Apply estimating poker for fast, relative estimating

- Estimating poker combines expert opinion, analogy, and disaggregation for quick but reliable estimates
- All members participate



?	1	2	3	5	8	13	20	40	100	∞

Steps	
1	Each estimator gets a deck of cards
2	Read user story
3	Estimators privately select cards
4	Cards are turned over
5	Discuss differences
6	Re-estimate

Source: Mike Cohn, Agile Estimating and Planning

Note: estimating poker is a form of the **Wideband Delphi** estimation method, a consensus-based technique for estimating effort.

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Estimation is a whole-team exercise

- Increases accuracy by including *all* perspectives
- Builds understanding
- Creates shared commitment



The whole team estimates Stories

Warning: *Estimation performed by a manager, architect, or select group negates these benefits*

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How much time to spend estimating

A little effort helps a lot. A lot of effort only helps a little.



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Activity: Estimate Stories

Prepare 10 min

Share 3 min

- **Step 1:** As a group, use estimating poker to estimate three Stories created in the previous activity.
- **Step 2:** Share with the class:
  - Where do you find challenges when engaged in Story estimation?
  - Are you as a team aligned around the combination of qualities that represent a Story point (volume, complexity, knowledge, uncertainty)?



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Tasking Stories

Some teams break their Stories into tasks and forecast them in hours to better understand their capacity and capabilities.

Team members discuss:

- Who would be the best person to accomplish it?
- Approximately how long would it take?
- What dependencies might it have to other Stories?





Tasking stories is an optional practice in SAFe

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Iteration Goals

Iteration goals provide clarity, commitment, and management information. They serve three purposes.



Align team members to a common purpose



Align teams to common Program Increment (PI) Objectives and manage dependencies



Provide transparency and management information

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Iteration goals: Examples

Software Example

Iteration Goals

1. Finalize and push last-name search and first-name morphology
2. Index 80% of remaining data
3. Other Stories:
   - Establish search replication validation protocol
   - Refactor artifact dictionary schema

Business Example

Iteration Goals

1. Roll out the GDPR incident report procedures
2. Prepare for external audit
3. Obtain approvals for financial report

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Commit to the Iteration goals

Team commitments are not just to the work. They are committed to other teams, the program, and the stakeholders.

A team meets its commitment:

By doing everything they said they would do,
- or -
in the event that it is not feasible, they must immediately raise the concern.

Commitment

Too much holding to a commitment can lead to burnout, inflexibility, and quality problems.



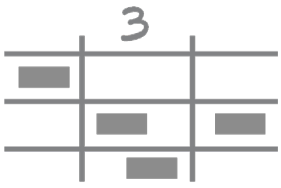
Adaptability

Too little commitment can lead to unpredictability and lack of focus on results.

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Iteration planning for Kanban teams

- Some teams have a more responsive nature to their work, such as maintenance teams and System Teams
- These teams may find less value in trying to plan the Iteration in detail, and will tend towards Kanban
- Kanban teams still publish Iteration goals, and integrate with other teams continuously or on cadence
- They commit to the goals as well as service level agreements (response time) for incoming work based on their known historical lead time
- They participate in PI Planning, System Demos and Inspect and Adapt like all other teams



Kanban

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Video: Implementing Kanban

Duration: 3 min



https://bit.ly/Video-ImplementingKanban

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Action Plan: Planning the Iteration

Prepare: 5 min

Share: 3 min

- **Step 1:** As a group, brainstorm one to three actions you could take to improve in any areas related to this lesson
- **Step 2:** Individually write down at least one improvement item
- **Step 3**: Share one item you discussed as a team and one item you individually wrote in your Action Plan



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Planning the Iteration

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Lesson review

In this lesson you:

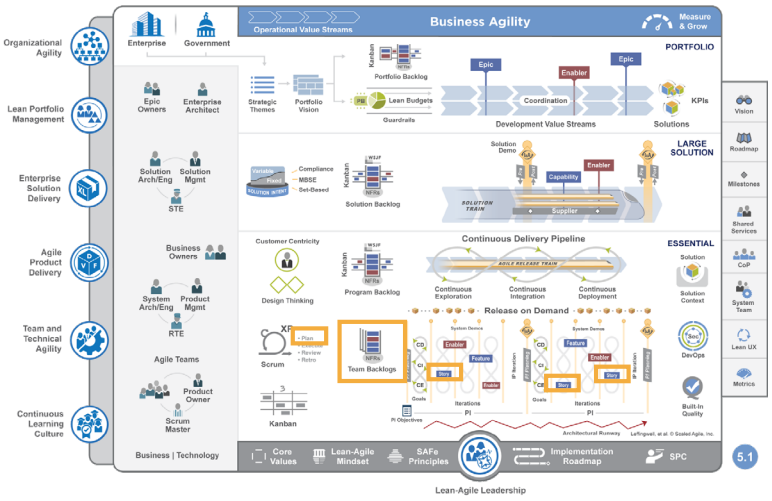
- Identified the qualities of well-written Features and Stories
- Prepared your backlog by breaking down Features into Stories
- Discussed the key considerations for sequencing work
- Applied estimation poker for fast, relative estimation
- Reviewed the process for planning the Iteration
- Explored Kanban as a team method for visualizing workflow

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Story" https://www.scaledagileframework.com/story/
- "Team Backlogs" https://www.scaledagileframework.com/team-backlog/
- "Iteration Planning" https://www.scaledagileframework.com/iteration-planning/



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Continue your SAFe journey with the following resources:

Watch the two videos in this playlist to learn more about the two different types of stories. https://bit.ly/Video-StoriesPlaylist	Access the “SAFe ART and Team Events” on the SAFe Community Platform for additional tools and guidance for preparing to facilitate SAFe events. https://bit.ly/Community-SAFeARTandTeamEvents
Download and read the “Story Writing and Splitting Guide” to learn what makes a good story and how to split stories. https://bit.ly/Community-StoryWritingGuide	Practice story splitting with your team using this SAFe Collaborate template, “Story Splitting on an Agile Team” https://bit.ly/Template-StorySplitting
Download the “SAFe Iteration Execution Toolkit (5.1)” for a set of tools and guides for facilitating significant Iteration events. https://bit.ly/Community-ToolkitsandTemplates	Download the “SAFe Remote ART Toolkit 5.1” for additional tools and guides for releasing value with distribute or remote teams. https://bit.ly/Community-ToolkitsandTemplates

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 4

Executing the Iteration

SAFe® Course - Attending this course gives students access to the SAFe® Practitioner exam and related preparation materials.

SP SAFe® 5

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Lesson Topics

4.1 Visualizing the flow of work

4.2 Measuring the flow of work

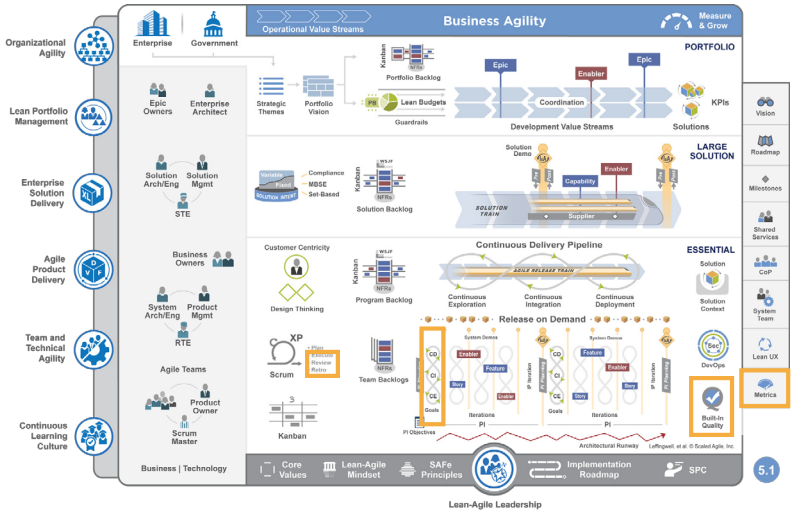
4.3 Building quality in

4.4 Continuously integrating, deploying, and releasing

4.5 Improving flow with communication and synchronization

4.6 Demonstrating value with the Iteration Review

4.7 Improving with the Iteration Retrospective



4-2

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Learning objectives

At the end of this lesson, you should be able to:

- Visualize your current flow of work
- Apply cumulative flow diagrams for measuring flow
- Identify the different aspects of Built-in Quality
- Recognize the importance of a Continuous Delivery Pipeline
- Describe the communication and synchronization benefits of the daily stand-up
- Explain how to run a successful backlog refinement session
- Describe the purpose of the Iteration Review and create a definition of done
- Participate in a retrospective and identify improvements

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4.1 Visualizing the flow of work

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Video: Designing your team's Kanban system

Duration 4 min



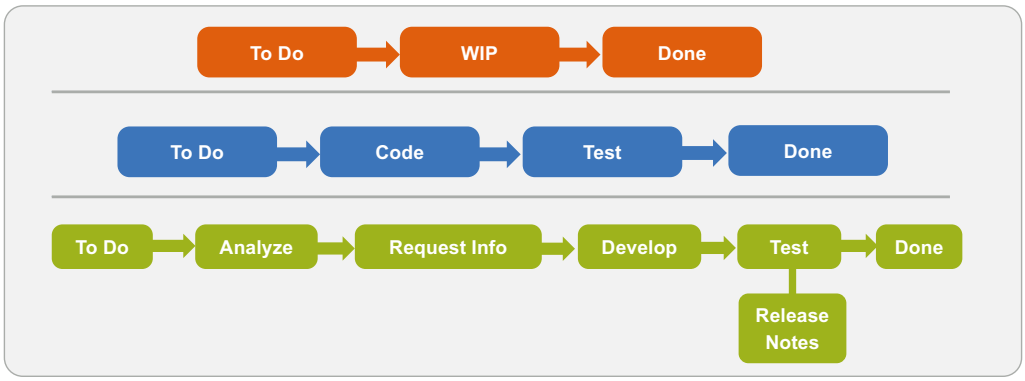
https://bit.ly/Video-DesignKanban

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Visualize the flow of work

- What is the flow of work for your team?
- What are the steps it takes to get a Story to done?

Example: Flow of work



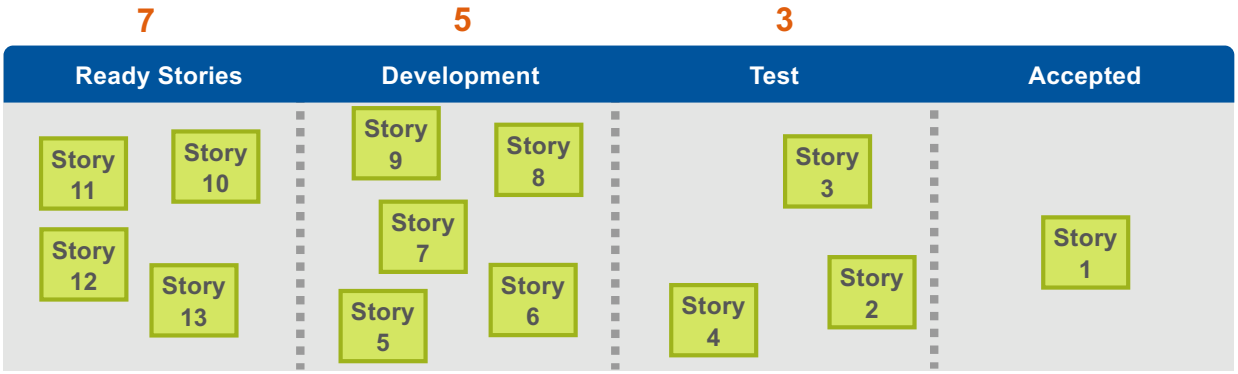
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Setting WIP limits

- WIP limits improve the flow of work
- Some steps have no WIP limits, while others serve as buffers and have minimum as well as maximum WIP



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Activity: Visualize the flow of work





- **Step 1:** Consider the Stories you estimated in the previous lesson
- **Step 2:** Define the steps you need to turn the Stories into value
- **Step 3:** As a group, build your current flow of work and assign WIP limits
- **Step 4:** Be prepared to share your flow of work with the class

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4.2 Measuring the flow of work

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SAFe defines six flow metrics



Metric	Description
Flow Distribution	The proportion of each backlog item type in the flow
Flow Velocity	Number of items completed in a given time
Flow Time	Time elapsed from when an item enters the workflow to when it is released to the customer
Flow Load	Total work-in-progress (across all steps of the flow)
Flow Efficiency	The portion of time backlog items are actively worked on to the total time elapsed
Flow Predictability	Overall planned vs. actual business value

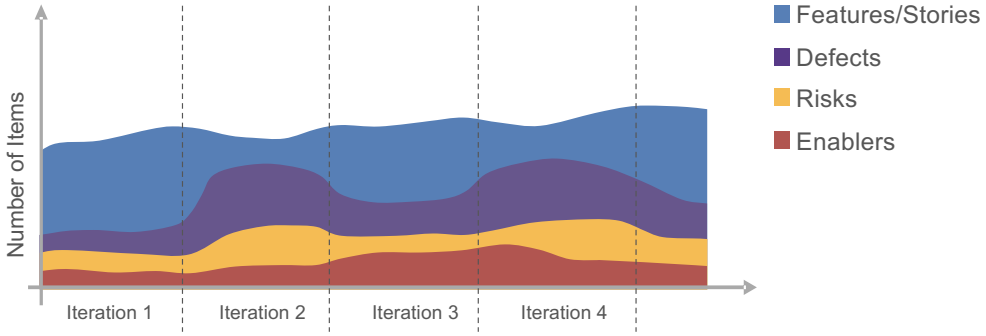
The first five metrics are from Mik Kersten's Flow Framework described in his book Project to Product

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Flow distribution ensures a healthy balance of work item types

What does it measure? Flow distribution measures the amount of each type of work in the system over time.

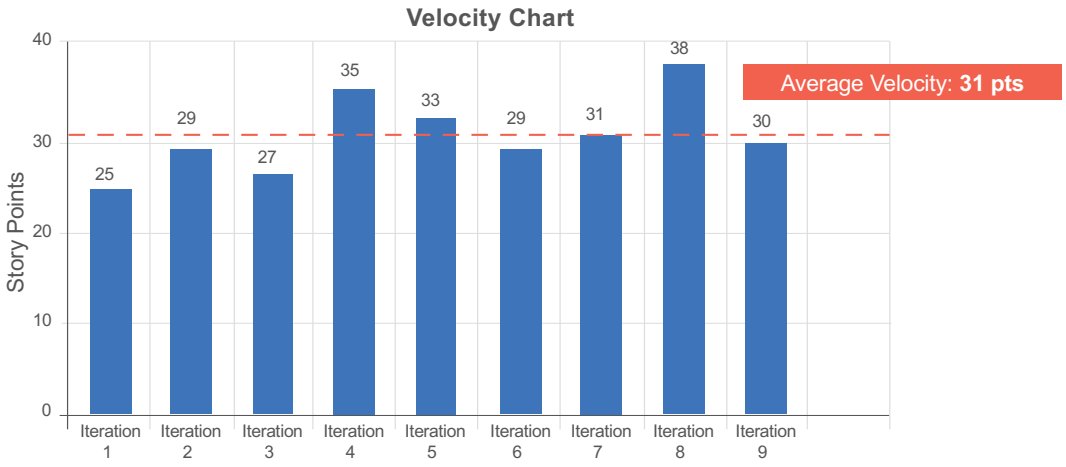


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Flow velocity tracks team performance each Iteration

What does it measure? Flow velocity measures the number of backlog items completed in a given timeframe



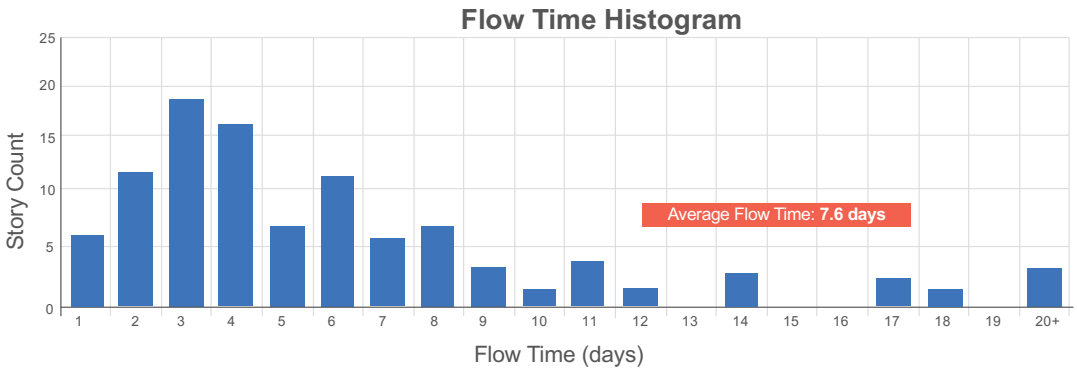
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Flow time ensures teams deliver value in the shortest possible time

What does it measure? Flow time measures the elapsed time from when an item enters the system (developer starts working) to the moment it is delivered to the customer.

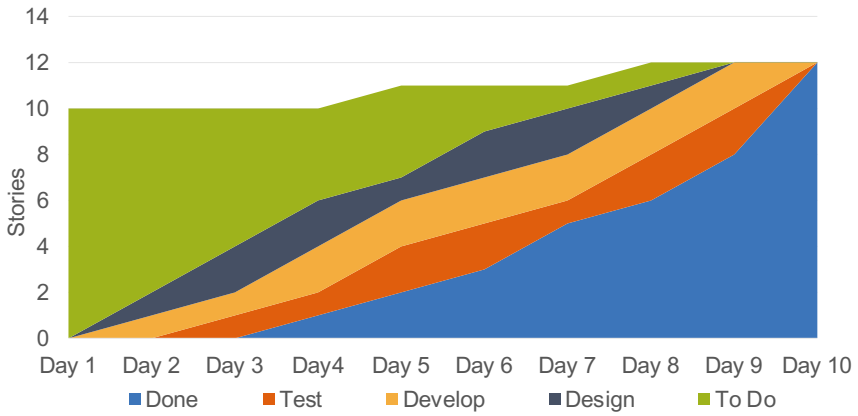


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Flow load is a leading indicator of excess work in process (WIP)

What does it measure? Flow load indicates how many items are currently in the system.

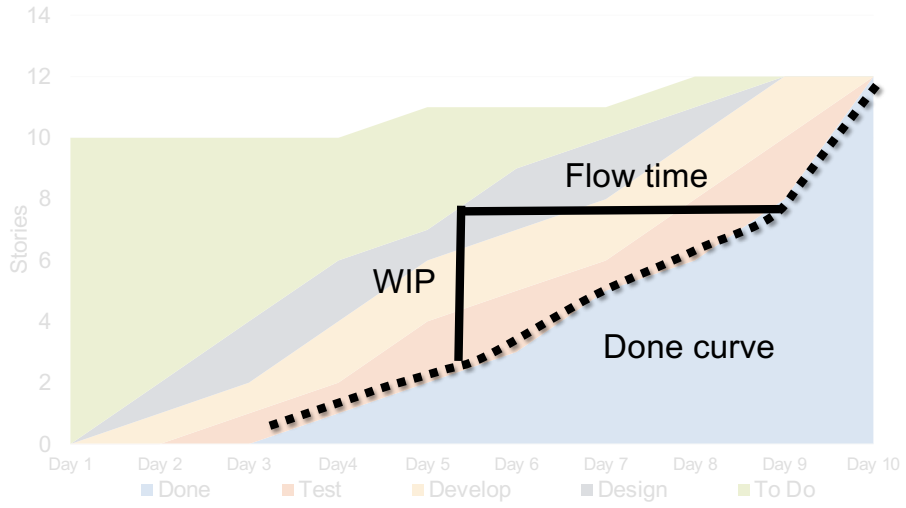


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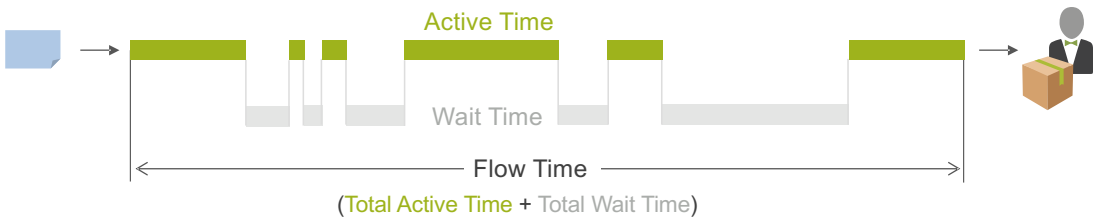
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What can you learn from a cumulative flow diagram (CFD)?



Flow efficiency highlights waste, bottlenecks, and delays in the system

What does it measure? Flow efficiency measures how much of the overall flow time is spent in value-added work activities vs. waiting between steps.



$$\text{Flow Efficiency} = \frac{\text{Total Active Time}}{\text{Flow Time}}$$

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4.3 Building quality in

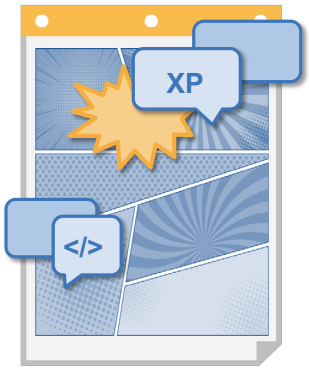
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Activity: Expressing “Build quality in”

Prepare 10 min

Present 5 min

- **Step 1:** In your group, discuss what building quality in means to you and your organization
- **Step 2:** Create a poster to reflect the different aspects of building quality in
- **Step 3:** Present your poster to the class



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Building quality in

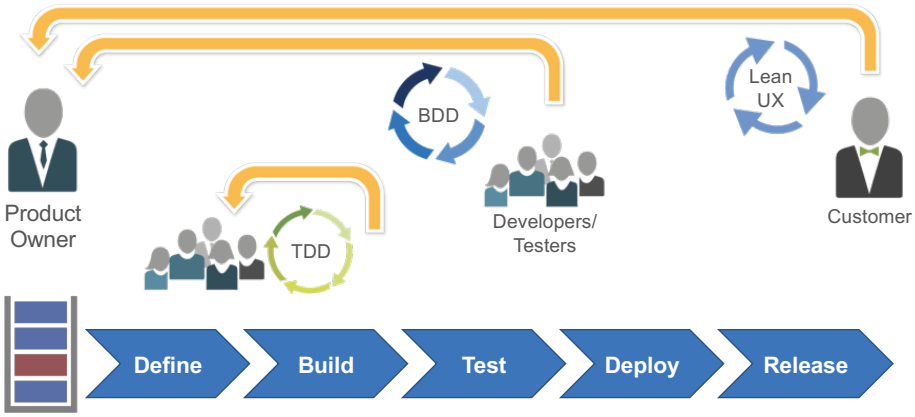
You can't scale crappy code (or hardware, or anything else). —Dean Leffingwell

- Ensures that every increment of the Solution reflects quality standards
- Is required for high, sustainable development velocity
- Agile quality practices apply to every team, whether business or technology:
  - Establish flow
  - Peer review and pairing
  - Collective ownership and standards
  - Automation
  - Definition of done

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Built-in Quality practices for software teams

They include software quality practices (most inspired by XP) like Agile testing, behavior-driven development, test-driven development, refactoring, code quality, and Agile architecture.

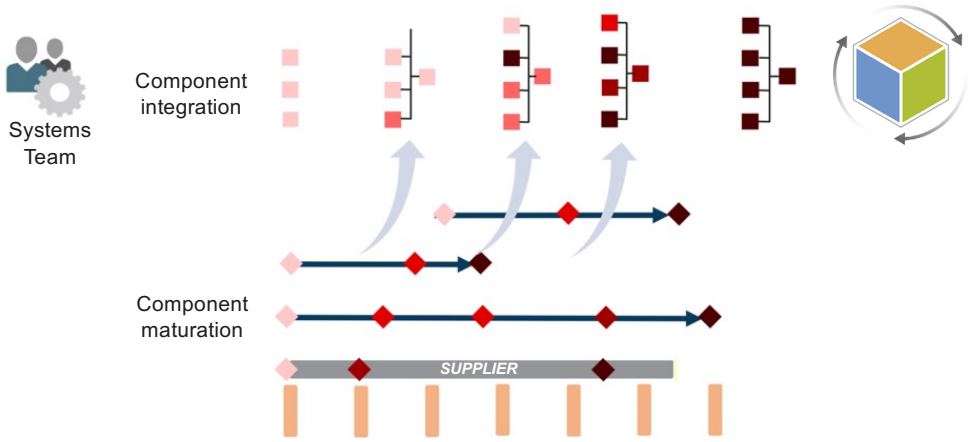


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Built-in Quality practices for hardware teams

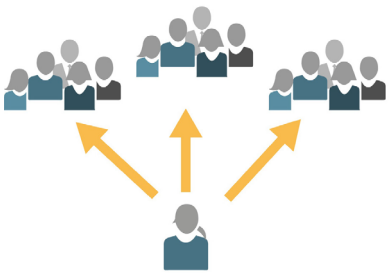
Support hardware quality with exploratory, early iterations, frequent system-level integration, design verification, Model-Based Systems Engineering (MBSE), and set-based design.



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Emergent design and intentional architecture

- Every team deserves to see the bigger picture
- Every team is empowered to design their part
- Emergent design - Teams grow the system design as user stories require
- Intentional architecture - Fosters team alignment and defines the Architectural Runway
- A balance between emergent design and intentional architecture is required for speed of development and maintainability



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Architectural Runway

Architectural Runway is existing code, hardware components, marketing branding guidelines, etc., that enable near-term business Features.

- Enablers build up the runway
- Features consume it
- Architectural Runway must be continuously maintained
- Use capacity allocation (a percentage of train's overall capacity in a PI) for Enablers that extend the runway

Example:
A single sign-on mechanism will enable sign-on in multiple applications



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Architectural Runway: Examples

Software Example

A new, fuzzy search algorithm will enable a variety of future Features that can accept potentially erroneous user input

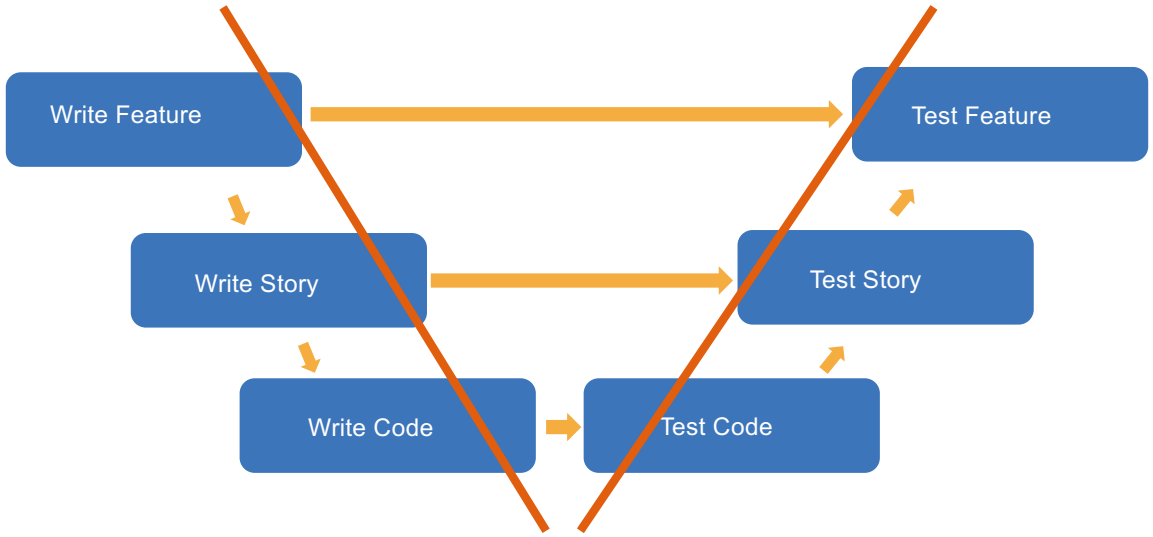
Business Example

HR Example: A job architecture and hiring strategy for Agile talent to enable the company's growth

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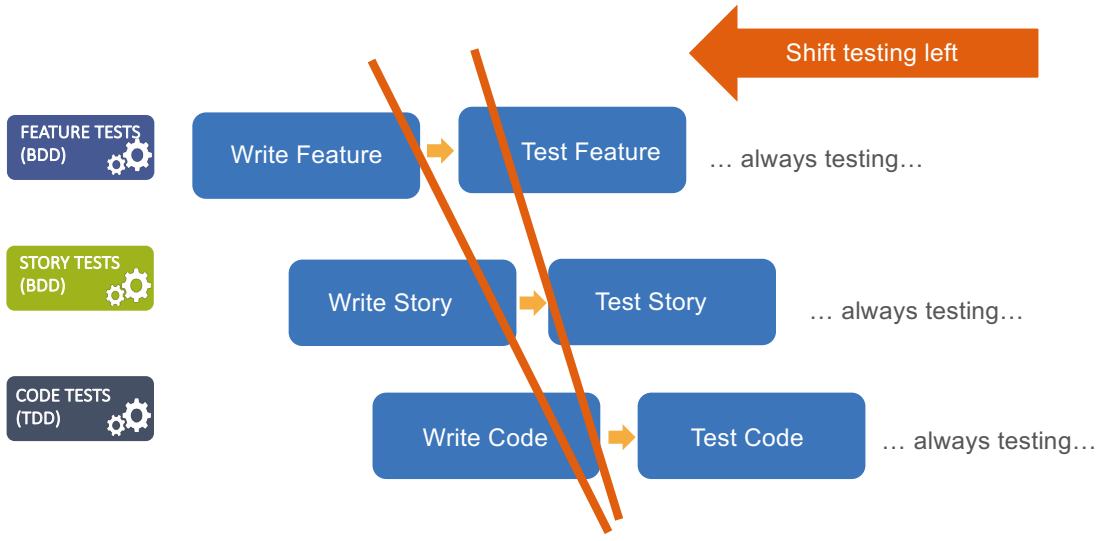
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Traditional testing (V-Model) delays feedback



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Shift testing left for fast and continuous feedback



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Test first naturally creates a pyramid of tests

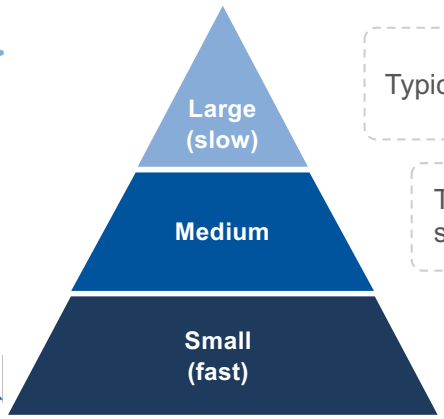
The test pyramid advocates a balanced portfolio of tests with many small, low-level, automated tests and fewer large, manual tests.

The test pyramid is about the speed of the test

Test

$$$

¢



Test Pyramid

Typical large end-to-end UI

Typical medium external services single UI

Typical small individual classes

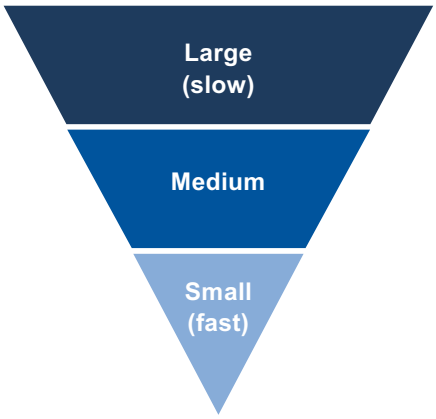
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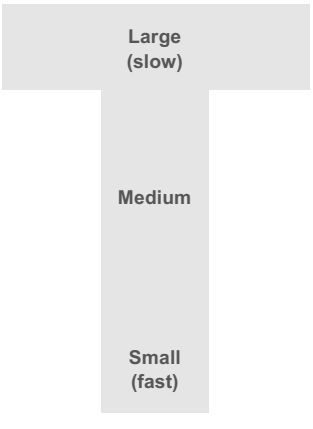
An inverted Test Pyramid is a test strategy anti-pattern

Slows development, delays feedback, encourages larger batches

Typically, UI

On a system not designed for testing





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4.4 Continuously integrating, deploying, and releasing

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Video: What is DevOps?

Duration 5 min



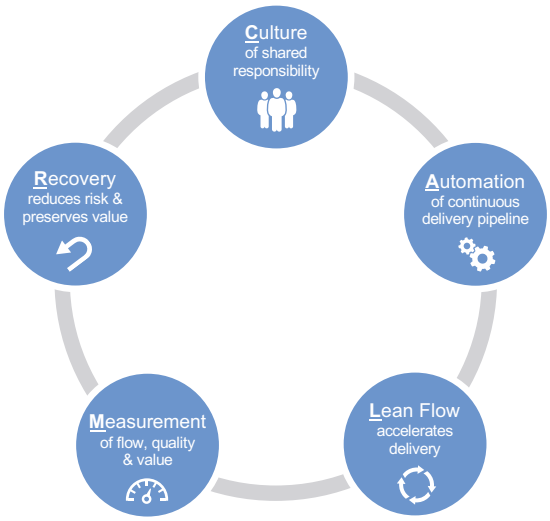
https://bit.ly/Video-WhatisDevOps

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A CALMR approach to DevOps

- **C**ulture - Establish a culture of shared responsibility for development, deployment, and operations.
- **A**utomation - Automate the Continuous Delivery Pipeline.
- **L**ean flow - Keep batch sizes small, limit WIP, and provide extreme visibility.
- **M**easurement - Measure the flow through the pipeline. Implement full-stack telemetry.
- **R**ecovery - Architect and enable low-risk releases. Establish fast recovery, fast reversion, and fast fix-forward.

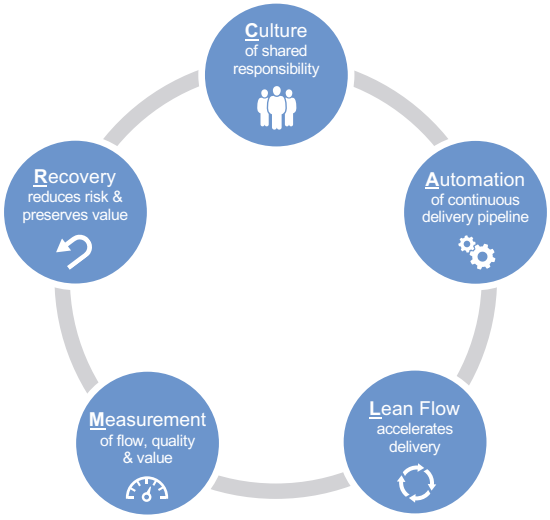


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Discussion: CALMR Approach

Duration 3 min

- **Step 1:** Discuss as a team:
  - What opportunities do you have in your current context to apply the concepts in the CALMR approach?
- **Step 2:** Be prepared to share with the class



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Video: Continuous Delivery Pipeline

Duration: 5 min



https://bit.ly/Video-CDP

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Discussion: Continuous delivery culture

Prepare: 5 min

Share: 3 min

- **Step 1:** As a group, discuss the following:
  - How is your culture or environment ready for continuous delivery?
  - What does “continuous” mean to you and your group?
- **Step 2:** Be prepared to share some insights with the class

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